Hyeongmin Choi

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Education

Chosun University — Gwangju, South Korea

B.S. in Artificial Intelligence Engineering (AI/SW School) — Mar. 2023 - Present (2nd Year)

- **Academic Details:** 3 years enrollment period (including military leave)
- **Relevant Coursework:** Artificial Intelligence, C++ Programming, Introduction to OSS (Open Source Software)

Skills

Core Competencies: C++ Programming, Data Structures, Real-Time Data Ingestion
AI-Assisted Development: Prompt Engineering, AI Agent Pairing, AI-Driven Prototyping (Vibe Coding)
Web Automation & Database: Flask Backend, Selenium Automation, Pandas Data Wrangling, SQLite/MySQL
Tools & Infrastructure: Docker, Docker Compose, Git, GitHub, Linux Environments

Projects

ASC_TLMTSYS (Motorsport Telemetry & Analysis System) — Sep. 2025 - Present

Role: Sole Developer (Personal Project)

- Developed a real-time 'F1 Pit Wall Control' telemetry dashboard serving dynamic vehicle physics data over a FastAPI (Port 8000) backend.
- Built a high-performance UDP socket listener (Port 8001) in Python/C++ to ingest live simulator packet streams at zero lag.
- Engineered live analytics tracking tire core/IMO temps (81.7°C), brake temps (417°C), suspension load, ERS battery, and DRS lock states.
- **Future Roadmap:** Scaling to hardware telemetry, designing sensor-acquisition boards on student-built formula race cars to ingest real track telemetry.

SCSS (Self Check-in Integrated System) v2.0 — Jan. 2025 - Apr. 2026

Role: Sole Developer / System Architect (Built during military service)

- Developed and operationalized a 24/7 unmanned self check-in kiosk interface on a Windows OS PC configured in Kiosk Mode.
- **Self-Taught & AI Collaboration:** Self-taught Python and network architectures, partnering with an AI agent to architect the program while serving as a Private First Class (PFC).
- Engineered an automated backend parser (Pandas & APScheduler) executing daily at 14:50 to process portal reservation Excel logs.
- Designed a modular plugin architecture since v1.0, enabling secondary utilities (e.g., cleaning checklist generator, network printing) to plug in seamlessly without altering core logic.
- Containerized the software stacks using Docker Compose, guaranteeing environment isolation and zero-downtime operations.

Honam-Univ.-POS (Real-Time POS & Mobile Order System) — May 2026 - Present

Role: Co-Developer / Backend & Database Architect (Festival Production Deployment)

- Co-developed and scaled a real-time POS and mobile ordering system for high-concurrency festival environments.
- Prevented SQLite database locks (*database is locked*) under heavy order traffic by implementing Write-Ahead Logging (WAL) and extending timeout to 30 seconds.
- Avoided lost update anomalies during peak order spikes via immediate transaction locking (*BEGIN IMMEDIATE*).
- Supported offline-capable, web-native mobile order interfaces utilizing Progressive Web App (PWA) configurations, serving real-time socket interfaces on Uvicorn (Port 3000).

Experience

System Administrator & Automation Developer — Jan. 2025 - Apr. 2026

Republic of Korea Army (Welfare Facility IT Automation)

- **SCSS:** Decreased check-in administrative bottlenecks by 90% via Windows OS Kiosk interface and Docker Compose.
- **Restaurant Booking System:** Developed a Flask/SQLAlchemy reservation app. Implemented security parameters (CSRF, secure session cookies, rotating log handlers) and underwent military security clearance reviews.

Telemetry Infrastructure Lead — Mar. 2024 - Present

Chosun University Formula SAE (Student Race Car Team)

• Joined to bridge mechanical engineering & embedded systems with AI/SW components, focusing on vehicle dynamics and telemetrics.
• Identified data logging instability on physical telemetry hardware and designed C++ real-time packet filtering modules.
• Implemented low-latency data pipelines for sensor preprocessing, shifting unstable legacy structures to stable telemetry platforms.

Career Goal

Aspiring **Race Data Engineer**. Leveraging solid software engineering foundations built from designing production-grade military IT automation systems to tackle new challenges in motorsport: developing low-latency telemetry pipelines (UDP/CAN) and optimizing real-time chassis dynamics.